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Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Time: 03 Hours

Code: 104404 ***Subject: Electromagnetic Theory*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only): **[14]**

(a) The divergence of the vector field is
(i) Vector
(ii) Scalar
(iii) Can't be both
(iv) None

(b) The unit of electric field intensity
(i) Volt
(ii) Volt per metre
(iii) Tesla
(iv) Metre

(c) In Gauss's Law the electric field is related to
(i) The charge density
(ii) The potential difference
(iii) The charge enclosed
(iv) The capacitance

(d) The electric field at a point due to an infinite sheet of charge is:
(i) $[\sigma] / [\varepsilon_0]$
(ii) $[\sigma] / [2\varepsilon_0]$
(iii) $[2\varepsilon_0] / [\sigma]$
(iv) 0

(e) The force on a charged particle moving in a magnetic field is maximum when the angle between the velocity and magnetic field is:
(i) 0°
(ii) 45°
(iii) 90°
(iv) 180°

(f) The energy density in magnetic field is given by
(i) $[\mu H^2] / [2]$
(ii) $H^2/2\mu$
(iii) 2
(iv) None

(g) The dot product of the vectors $3i-2j+5k$ and $-i+3j+2k$ is,
(i) -1
(ii) 1
(iii) 0
(iv) -3i

(h) The Pointing vector P is
(i) $E \times H$
(ii) $E \cdot H$
(iii) $1/2E \times H$
(iv) $(E \times H)^2$

(i) The Maxwell's equation $\nabla \cdot B = 0$ is due to
(i) Monopole does not exit
(ii) Coulomb's Law
(iii) Existence of monopoles
(iv) Ampere Circuital Law

(j) Curl of gradient of a vector is
(i) 0
(ii) Null Vector
(iii) Unity
(iv) Depends on the constants of the vector

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Q.2 Solve both questions : **[7+7=14]**

(a) Write the differential elements (dl, da, dv) in both Cartesian, cylindrical co-ordinate system.

(b) State divergence theorem. What will be divergence to position vector?

Q.3 Solve both questions : **[7+7=14]**

(a) Given the two points, C(−3,2,1) and $D(r=5,\theta=20^{\circ},\varphi=-70^{\circ})$. find: (i) The spherical coordinates of C (ii) The Cartesian coordinates of D

(b) Find the divergence of $A= 3x^2a_x+ 5x^2y^2a_y+ xyz^3a_z$ where a_x, a_y and a_z are unit vectors in cartesian coordinates at point (1,1,1)

Q.4 Solve both questions : **[7+7=14]**

(a) State and derive Coulomb's law. Write coulomb's law in vector forms.

(b) Derive an expression for intensity of electric field at a point distant r from a point charge.

Q.5 Solve both questions : **[7+7=14]**

(a) An infinite long line charge of uniform density ρ_L C/cm is situated along the z-axis. Obtain electric field intensity due to this charge using Gauss's law.

(b) Derive energy density in electrostatic field.

Q.6 Solve both questions : **[7+7=14]**

(a) Four 3pC charges are at the corners of a 1-m square. The two charges at the left side of the square are positive. The two charges on the right side are negative. Find the field E at the centre of the square, $\varepsilon_r=1$

(b) What do you understand by capacitance of a capacitor? Deduce and expression for the capacitance of a parallel plate capacitor. How will it be modified when the gaps between the plates is filled with a dielectric?

Q.7 Solve both questions : **[7+7=14]**

(a) Show that the Faraday's law of electromagnetic induction can be expressed as $\nabla\times E = -\partial B/\partial t$. Write down its integral form.

(b) Explain the concept of displacement current and show how it led to the modification of the Ampere's law.

Q.8 Solve both questions : **[7+7=14]**

(a) Discuss reflection of plane electromagnetic wave incident normally on a perfect dielectric and obtain expressions for the two reflection coefficients of electric and magnetic fields.

(b) Define Poynting vector. Mention any two properties of uniform plane wave.

Q.9 Write short notes on any two of the following:- **[14]**

(a) Biot-Savart's Law
(b) Magnetic field boundary condition for isotropic medium
(c) Skin Effect
(d) Spherical capacitance

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